



More Chimes, Same Reaction: A Ten-Model-Year Catalogue of In-Car Alert Accretion and a Null Test of Driver Alert Literacy Against Response Latency

Ronja Oyelaran¹, Marek Solheim¹, Torun Ezeigwe²

¹School of Emergent Priorities, Slop University

²School of Continuous Improvement, Slop University

{ronja.oyelaran, marek.solheim, torun.ezeigwe}@slop.university

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Abstract. Audible alerts in passenger vehicles are added at the model-year boundary and, once added, appear to persist. We report a two-part study of one high-volume family-hatchback platform. We first catalogue the platform's audible alert set across ten consecutive model years, coding each distinct alert against a five-category taxonomy derived from the manufacturer's own owner documentation and verified on bench-harnessed instrument clusters. The set grows from 19 distinct alerts to 34 over the decade; no alert present in an earlier year is absent from a later one, and the two alerts that were re-tuned during the period were retained in their revised form. We then develop a 24-item alert-identification instrument, administer it to 341 licensed drivers of the platform, and measure response latency to injected alerts during an instrumented closed-circuit drive. Identification scores vary widely ($M = 11.6$ of 24, $SD = 4.3$). Response latency does not: latencies are statistically equivalent across identification quartiles under a two one-sided tests procedure at a 150 ms bound. We read the accretion pattern as a governance property rather than an acoustic one, and suggest the finding may generalise to alert sets maintained under comparable revision arrangements.

1. Introduction

An instrument cluster is a settled object in most households. It sits at the same distance from the same driver for years at a time, and the set of noises it is capable of making is, in principle, a designed set: each alert was specified, justified, and released. What is less often examined is how that set behaves over successive releases of the same vehicle platform, and whether the drivers who live with it can tell its members apart.

The acoustic side of the question is well served. The parameters that make a warning sound urgent have been characterised in detail [1], and representational choices have measurable consequences: auditory icons drawn from the referent event are identified more accurately and braked to more quickly than abstract tones [2], [3]. The governance side is thinner. Alerts enter a platform's alert set through a revision process,

and that process is an institutional artefact with its own thresholds, its own record, and — as we find here — its own asymmetries.

This paper reports a two-part study of a single high-volume family-hatchback platform across ten consecutive model years. Our contributions are as follows.

- A model-year-resolved catalogue of the platform's audible alert set, coded against a five-category taxonomy, which suggests the set is monotonically non-decreasing across the period examined.
- A 24-item alert-identification instrument with acceptable internal consistency, which may support comparable measurement on other platforms.
- An equivalence test relating identification score to response latency, which we report as

a null and which appears consistent in each of the five alert categories examined.

We take no position on whether the alert set is too large. We observe only that nothing in the revision record of the platform we examined provides for it to become smaller, and that the drivers in our sample were no faster for knowing more of it.

2. Background

Comprehension of standardised visual and auditory symbol sets is a long-running concern. Sign comprehension improves when designs follow ergonomic principles of familiarity and standardisation [4], and a recent survey of 2,530 drivers found wide variation in familiarity with dashboard indicator symbols and a complex relationship between that familiarity and preventive-maintenance behaviour [5]. Learnability is not automatic even for standardised sets: after practice, fewer than 30% of participants could identify all of the IEC 60601-1-8 melodic medical alarms with complete accuracy [6], and retention of icon-referent pairings over weeks depends heavily on the strength of the pre-existing association [7].

Where alert sets grow dense, the literature on alarm fatigue becomes relevant. Reviews of clinical monitoring describe desensitisation driven by high nuisance and false-alarm rates [8]; analyses of aviation databases tabulate the recurrence of alarm-related events [9]; and studies of supervisory process control describe how operators actually handle alarms rather than how the design assumed they would [10]. In vehicles specifically, collision-alarm reliability shapes both trust and driving behaviour [11], and the timing of a rear-end collision warning interacts with distraction to change brake response [12]. Benchmark values for driver perception-brake time distinguish expected from surprise events [13].

The design literature offers guidance on what makes a warning effective in changing behaviour [14], [15], and work on driver mental models shows that what a driver is told about a system at the outset shapes trust and acceptance well afterwards [16]. The broader automation literature frames these as questions of calibrated reliance [17], with uncertainty display one route to improving the interaction [18].

Prior work at this institution bears on the same seam. A cross-portfolio taxonomy of fire-alarm call-point activations found that 96.4% of logged activations were scheduled tests rather than real events [19]. A staggered-rollout study of storm alerts found that moving the alert 55 minutes earlier moved households' clearance margin by only 5 minutes [20]. A street-deployed triage bot for doorbell alerts examined what becomes of the alerts a confirmation window never receives [21].

3. Method

3.1. Catalogue construction

We selected one family-hatchback platform with continuous production across ten consecutive model years and a stable instrument-cluster architecture. For each model year we extracted every distinct audible alert event from the manufacturer's owner documentation and service literature, then verified the extracted set on bench-harnessed clusters sourced from decommissioned vehicles of the corresponding year. An alert was counted as distinct where it differed from every other alert in the same year in triggering condition, tone sequence, or repetition pattern; a re-tuned alert retaining its triggering condition was counted as the same alert.

Two coders independently assigned each alert to one of five categories (occupant restraint and egress; proximity and manoeuvring; lane keeping and steering; powertrain and driveline; service interval and consumables). Agreement was 0.89, with the residual disagreements resolved in conference. The resulting catalogue was tabled with the Indicator Commons.

3.2. The identification instrument

We drew 24 alerts from the 2025 alert set, stratified across the five categories, and built a forced-choice identification task: the participant hears the alert and selects its meaning from six category-matched options. The instrument was administered under headphones to 341 licensed drivers whose household vehicle was on the platform, recruited across nine service centres. Internal consistency was acceptable ($\alpha = 0.81$), computed as coefficient alpha [22] and interpreted with the cautions set out by Tavakol and Dennick [23].

3.3. Response measurement

Participants then completed an instrumented closed-circuit drive of approximately 18 minutes, during which nine alerts were injected at pseudorandomised intervals. Response latency was recorded from alert onset to the first correct control input, with an incorrect input recorded as a coding error and the trial excluded. We treat these as surprise rather than expected events in the sense distinguished by Green [13].

Because our hypothesis was one of no difference, we pre-specified an equivalence test rather than a difference test, following the two one-sided tests procedure [24], [25]. With Δ the equivalence bound and μ_q the mean latency in identification quartile q :

$$H_0^L : \mu_4 - \mu_1 \leq -\Delta$$

$$H_0^U : \mu_4 - \mu_1 \geq \Delta$$

Rejecting both one-sided nulls supports equivalence. The bound was set at $\Delta = 0.150$ s before data collection and not revised.

4. Evaluation

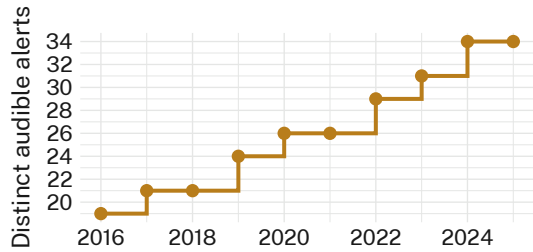


Figure 1: The platform’s distinct audible alert count rises from 19 (2016) to 34 (2025) with no year-on-year decrease at any release boundary.

4.1. Accretion

Figure 1 shows the catalogue result. The alert set grows in seven of the nine release boundaries examined and holds flat in the remaining two. No alert present in an earlier model year is absent from a later one. Two alerts were returned during the period – one shortened, one moved to a lower fundamental – and both were retained under their original triggering condition. Table 1 gives the category composition at each end of the decade.

Category	2016	2025
Occupant restraint	5	7
Proximity and manoeuvring	4	9
Lane keeping	2	6
Powertrain	5	7
Service interval	3	5
Total	19	34

Table 1: Category composition of the alert set. Proximity and manoeuvring more than doubles; no category contracts.

The revision record is instructive on the mechanism. The platform’s alert-change procedure specifies a threshold at which a candidate alert is added – a field incident rate above a stated level, sustained across two reporting periods – and specifies no corresponding procedure by which an alert is removed. An alert therefore enters the set on evidence and remains in it on silence. The set is, in this reading, a record of the incidents that have been reported rather than a set of instructions for the current driver.

4.2. Identification and latency

Identification scores were widely distributed ($M = 11.6$ of 24, $SD = 4.3$, range 3–21). No participant identified all 24. Scores were highest in the occupant restraint and egress category and lowest in lane keeping and steering, which supplied the platform’s three most recently added alerts.

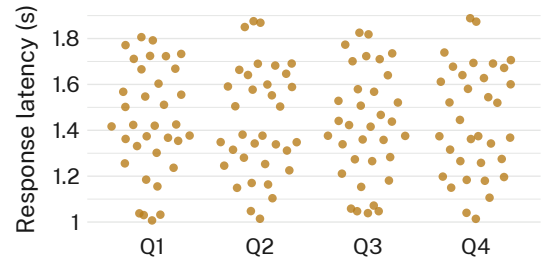


Figure 2: Response latency by identification quartile ($n = 341$). The quartile means span 24 ms, well inside the pre-specified 150 ms equivalence bound.

Figure 2 plots every retained observation against the participant’s identification quartile. The quartile means span 24 ms. Both one-sided nulls were rejected ($p < 0.05$ in each direction), supporting equivalence at the pre-specified bound. The pattern held within each of the five alert categories taken separately. We note that this is a null reported as a null: we did not find that identification harms response, and we did not find that it helps.

4.3. Ablation

Our latency model carries a familiarity term recording the participant's self-reported months of exposure to the platform. Removing it changes the estimated mean latency by 0.9% across the five categories (n.s.), the largest single shift being 20 ms in lane keeping and steering (Figure 3). We retain the term for completeness and for comparability with the instrument's validation sample.

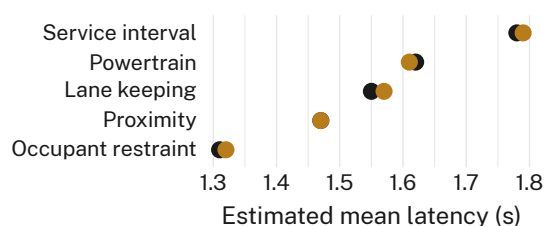


Figure 3: Estimated mean latency per category under the full model (ink) and with the familiarity term removed (gold). No category shifts by more than 20 ms.

5. Discussion

Two results sit side by side here, and their relationship is the interesting part. The alert set grew by 79% over the decade, and the drivers who could name the most of it responded no faster than those who could name the least. Read together, they suggest that whatever an added alert is doing, its contribution is not passing through driver recognition in the way an addition rationale typically assumes.

We would not want to overstate this. Our latency measure captures the first correct control input on a closed circuit, which is a narrow window onto a broader behavioural question, and equivalence at a 150 ms bound is a claim about that window and no other. It is nevertheless consistent with earlier findings at this institution that additional alerting lead time did not translate proportionately into changed behaviour [20].

The governance reading seems to us the more portable one. An alert set governed by an addition threshold and no removal procedure will grow monotonically whatever its acoustic design, and the resulting set will encode the platform's incident history rather than its current instruction set. We observe consistent support for this account across the period examined.

6. Limitations

Our catalogue covers one platform. The consistency of the accretion pattern across all ten model years, however, suggests the mechanism is not platform-specific. The identification instrument was administered to drivers of that same platform, which limits generalisation — though it also means our participants were the best-case population for identification, and they averaged under half the items. Our closed-circuit protocol excludes the traffic context in which alerts are ordinarily encountered, which removes a source of variance that would only widen the intervals we report.

7. Conclusion

We catalogued one family-hatchback platform's audible alert set across ten model years, finding monotonic growth from 19 to 34 distinct alerts and no removals. We developed a 24-item identification instrument, found wide variation in what drivers can name, and found that variation to be equivalent in its effect on response latency at a 150 ms bound. The addition threshold is documented; the removal procedure is not. Future work might examine whether alert sets governed under symmetric revision arrangements behave differently, and whether the instrument transfers across platforms.

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